MAT SCI 298 – Machine Learning for Materials Science
Winter 2025 – Prof. Daniel Schwalbe-Koda

Course Contents and Learning Objectives

Description: Lecture, three hours; computer laboratory, one hour. Introduction to data science and machine learning applied to materials science. Topics include: data analysis and visualizations; databases and materials informatics; classical methods for unsupervised and supervised learning; deep learning; materials discovery, selection, and optimization pipelines; uncertainty, explainability, and frontiers of ML-based materials science.

Course Pre-requisites (undergraduate level): COM SCI 31, MATH 31A, MATH 33A, and MAT SCI 104 (or equivalent courses). Proficiency with the Python programming language.

Knowledge Pre-requisites:

- Basic materials science (i.e., understanding structures, structure-property relationships).
- Proficiency with the Python programming language.
- Understanding of calculus and linear algebra.

References for introductory Machine Learning (ML):

- A. Burkov. *The Hundred-Page Machine Learning Book.* Andriy Burkov, 2019.
  - Available at: https://themlbook.com/
- C. Bishop. *Pattern Recognition and Machine Learning*. Springer Science+Business Media, LLC, 2006.
  - Available at: https://www.microsoft.com/en-us/research/uploads/prod/2006/01/Bishop-Pattern-Recognition-and-Machine-Learning-2006.pdf
- M. Nielsen. *Neural Networks and Deep Learning.*
  - Available at: http://neuralnetworksanddeeplearning.com/
- I. Goodfellow, Y. Bengio, and A. Courville. Deep Learning. MIT Press, 2016.
  - Available at: https://www.deeplearningbook.org/

References for ML + Materials Science or Chemistry:

- A. White. Deep Learning for Molecules and Materials. *Living Journal of Computational Molecular Science* **3** (1), 1499 (2021).
  - Available at: https://dmol.pub/index.html
- K. T. Schütt et al (eds). Machine Learning Meets Quantum Physics. *Lecture Notes in Physics* **968**. Springer, 2020.
  - Available at: https://link.springer.com/book/10.1007/978-3-030-40245-7
- A. R. Leach, V. J. Gillet. *An Introduction to Chemoinformatics*. Springer, 2007.
  - Available at: https://link.springer.com/book/10.1007/978-1-4020-6291-9
- K. Rajan (ed). *Informatics for Materials Science and Engineering*. Butterworth-Heinemann, 2013.
  - Available at: https://doi.org/10.1016/C2011-0-06098-2
- Machine Learning at the Atomic Scale. *Chemical Reviews* **121** (16), 9719 (2021).
  - Available at: https://pubs.acs.org/toc/chreay/121/16

Learning objectives:

- Understand the connection between materials science and data science, representations, and specific issues related to materials data.
- Answer scientific questions within materials science using a data-driven approach.
- Write computer code to create structure-property relationships, materials optimization algorithms, and materials selection strategies.
- Describe and provide critical assessment on data-driven pipelines in materials science and their context.
- Use classical machine learning and deep learning methods to create models connecting materials and molecular science.

Course Logistics

Role	Name	E-mail	Office Hours
Instructor	Daniel Schwalbe-Koda	dskoda@ucla.edu	Thu. 4-5pm @ 3532E Boelter Hall

Lectures and Laboratory:

- Lectures: Tuesday and Thursdays, 2:00pm - 3:50pm, Boelter Hall 5272
- Every lecture, we will have code demonstrations as part of a computer laboratory. Participating in lectures and laboratory, as measured by responses to the quizzes in the lecture, is part of the grading.
- During the lectures, we will cover some theoretical background in machine learning, solve some conceptual and programming problems, and write computer code.
- *Please bring your computer to the lectures*. We will have programming demonstrations and laboratories, which you will take part on.

BruinLearn link: https://bruinlearn.ucla.edu/courses/200937

- Lecture slides and relevant course material will be posted on BruinLearn prior to the lecture.
- Homeworks will be posted on BruinLearn on the day they are assigned.
- All homeworks should be submitted directly to BruinLearn

Communication:

- Announcements, class content, and more will be posted on BruinLearn.
- If you have questions that may be of general interest, please post it on the "Discussions" section of our BruinLearn page. This will help others view the responses.
- If e-mail communication is used with the instructor, try to tag MSE298 in the beginning of the subject line, e.g., "[MSE298] Question about…". This will help the instructor to see your e-mail and prioritize the response. Typically, you can expect a response within two business days.
- In general, in-person communication is preferred, especially if you have questions regarding debugging, assignments, and discussions that require explanation.
- The instructor may send occasional e-mails with reminders or information about the course. These e-mails will be sent through BruinLearn as announcements.

Tentative schedule:

#	Date		Topic	HW	Lab	Notes
1	25-01-07	T	Introduction; data science & visualization for materials science	1		
2	25-01-09	R	Materials datasets & databases; materials informatics; data exploration	1	1	
3	25-01-14	T	Data representation; feature engineering; encoding physics in data	1		
4	25-01-16	R	Unsupervised learning; dimensionality reduction; clustering; applications	1	2	
5	25-01-21	T	Basic regression/classification methods; evaluation; regularization	2		
6	25-01-23	R	Ensemble learning; kernel methods; structure-property relationships	2	3	JC prop. due
7	25-01-28	T	Gaussian process regression; property optimization; materials selection	2		
8	25-01-30	R	Deep learning and NNs; training methods; applications to materials	2	4	
9	25-02-04	T	**Journal Club**			
10	25-02-06	R	**Journal Club**			
11	25-02-11	T	Convolutional neural networks; microscopy data; mesoscale applications	3	5	proj. prop. due
12	25-02-13	R	Recurrent neural networks; sequence-based chemical data; literature data;	3	6	
13	25-02-18	T	Graph neural networks; materials property prediction; ML force fields	3	7	
14	25-02-20	R	Uncertainty quantification; robustness; explainability and materials science	4		
15	25-02-25	T	Generative models: VAEs, GANs, autoregressive approaches; LLMs	4	8	
16	25-02-27	R	Materials discovery pipelines; high-throughput screening; calibration	4		
17	25-03-04	T	**Project preparations & discussions**			Over Zoom (?)
18	25-03-06	R	Frontiers of materials-based ML; RL; lab automation			
19	25-03-11	T	**Project presentations**			
20	25-03-13	R	**Project presentations**			

Reading recommendations are provided directly on BruinLearn.

Learning Assessment

Grading:

Type	Quizzes	Participation	Homework	Journal Club	Project	Extra
% grade	10%	10%	25%	20%	35%	2%

- The final letter grade will be obtained by obtaining the average of the grades weighted as shown above.
- Final grades will not be rounded at the end of the course. Once the bins for letter grades are established, we unfortunately cannot round up the grades just because they land close to a threshold. The extra credit is your best chance to get this "round-up" at the end of the course.

Quizzes:

- Each lecture will contain quizzes for you to test your conceptual understanding of the previous lecture.
- The quizzes are multiple-choice or single-word answer that will be automatically recorded on BruinLearn. The quizzes compose 10% of your grade.
- The quizzes very often will contain open-ended or multiple-choice questions. These questions are incentivized for you to obtain a good understanding of the topic and be able to debate the models, strategies, and outcomes of the analysis.
- Out of the 15 lectures, you are required to respond the quizzes for 10 of them to obtain full credit for the quizzes.

Participation:

- This course contains credit for: engaging in discussions on BruinLearn; sharing relevant papers, methods, packages, videos, and code with the class through the discussion section on BruinLearn; and similar activities. The resources should be related to the topics of interest, with a preference for resources in chemistry/materials science.
- Participation is measured by your level of engagement in the course, as measured by your engagement in discussions and content sharing on BruinLearn.
- We request that you share at least **three** papers or resources (e.g., GitHub repositories, blog posts containing updates from companies etc.) of your interest throughout the course on BruinLearn to obtain full credits. These resources should not overlap with those from other students to maximize the diversity of ideas we are exposed to.
- Sharing resources should be performed using the "Discussion" section of BruinLearn. Instead of simply adding a link to a relevant paper, it is important to connect the shared resource with the topics seen in the lecture, and summarize important points of the paper. Do not simply paste a link to a paper in the Discussion thread – this will not be counted as participation.
- Another way to participate is to engage in questions and propose active discussions on BruinLearn. You can answer your colleagues' questions too, or provide your perspective. Imagine that this is similar to a forum where questions and answers are exchanged about a certain topic.
- Participation during lectures in the form of questions, discussions, and so on, is optional. Not engaging during the lectures does **not** negatively affect your participation grade. Nevertheless, you are encouraged to ask questions and discuss the subject, examples, and applications. This course is designed to favor discussions, critical thinking, and more.

Homeworks (HWs):

- Designed to practice the concepts under study. We have a total of (4 + 1) homeworks in this course, organized by topic.
- The four main HWs compose 25% of your final grade to promote continuous learning.
- HW0 (zero) is a pre-requisite for the course. If you cannot complete homework 0, we will **not** let you proceed with the course. This is because the entire course is based on knowledge of programming, and not knowing basic programming in Python will undermine your chances of following this course. Unfortunately, we do not have enough time to cover basic programming exercises in this course, so please take this into account when attending this course.
- HWs are roughly organized by topic to be coherent. They are assigned approximately every two weeks.
- Deadlines for the homeworks are long so you have time to do all problems, ask questions to the instructor, and more. Due dates are often one week and a half after they are assigned.
    - Please note that the instructor may be unable to answer last-minute questions for the homework before the deadline, especially when it comes to debugging.
    - Software development may take time and lead to unexpected errors, so plan ahead to solve the homework!
- Late homework policy:
    - You are allowed to have up to one late submission in homeworks without penalty

  - A homework is considered late if it is submitted more than one hour after the deadline.
  - If the homework is submitted between 1 h and 24 h after the deadline, we will use one of your late submissions.
  - Any homework submitted more than 24 h after the deadline is automatically graded as zero, regardless of how many late submissions you have available.
  - An extension can be granted by the instructor on a case-by-case basis.
  - Extensions are only provided with advance notice.

Journal club:

- Tentatively scheduled for February 2 and 4, 2025, during the lecture time.
- The journal club is a moment where students are requested to present a critical review of a paper or method from the literature. The critical review should cover the context of a given paper, the methods developed, dataset utilized, evaluation criteria, and so on.
- Students will have to debate the merits and shortcomings of the paper in a respectful way and based on evidence. Comparison against other methods in the literature is encouraged, as is a review of the field.
- Students will have to give a 10-min presentation (with slides) during the lecture to describe the paper. The paper will be debated among students after the presentation.
- Students are allowed to discuss the papers and presentations with others ahead of the talk.
- Graduate students (Masters and Doctoral levels) should submit individual work for their journal club presentations.
- Undergraduate students can perform the journal club assignment in teams of up to 2 people.
- Detailed instructions on the journal club assignment are provided in a separate document.

Final project:

- The final project is a small, research-like activity investigating a problem in the intersection of materials sciences and machine learning, worth up to 35% of your total grade.
- The grade will be composed by two parts: 5% for the proposal; 20% for the final project concept and execution; and 10% for the final presentation.
- The proposal will be due a month prior to the final project presentation. The proposal is a short text describing the work you will explore along with the relevant references.
- As presentation, students will have to give a 10-min presentation (with slides) to describe their work. The video will be shown to the class and debated among students after the presentation. Students should make a presentation similar to a conference talk, with a rough structure of 2 mins of introduction, 6-7 mins to describe the methods, results and discussion, and 1 min of conclusions and takeaways.
- The final project can be performed alone or in teams of two. Larger groups require permission from the instructor, justification on how the larger project justifies the work, and integration plan.
- Detailed instructions on the final project are provided in a separate document.

Extra:

- Outstanding participation during quizzes and on BruinLearn will be rewarded with an additional 1.5% of your total grade
  - Responding at least 13 quizzes out of the 15 will add an extra 0.75% to your grade.

    - Sharing more than 6 resources throughout the quarter will add an extra 0.75% to your grade.
- Filling out the course evaluation at the end of the quarter will add 0.5% of your total grade.
- The course evaluation is totally anonymous. It is a **really** important feedback you can provide to the school, the department, and the instructor.
- The extra credits are another chance to "round up" your final grade and improve your letter grade.

Solutions:

- Students will be provided with feedback for quizzes during the lectures.
- Feedback for homeworks will be provided after the deadlines.

Accommodations:

- In principle, this course does not use traditional exams administered by the Center for Accessible Education (CAE). The evaluation methods will be in the form of homeworks, recorded presentations, virtual participation, and a final project.
- Nevertheless, if you have approved accommodations with the CAE, please talk to the instructor as soon as possible on whether they apply to this evaluation criteria and discuss how to satisfy them.
- For more information on accommodations, please visit the CAE website, visit the CAE at A255 Murphy Hall, or contact CAE by phone at (310) 825-1501.

Ethics & Agreements

- Students are expected to comply with UCLA's Student Conduct Code.
    - Non-compliance with the code of conduct through practices such as cheating or plagiarism can justify an investigation regarding academic misconduct from UCLA's Office of the Dean of Students.
- Artificial intelligence (AI)-based or generative models can be used for learning/discussion purposes, revision of textual correction, or help with writing computer code.
    - This course is focused on AI, so it is natural to encourage the use of these tools throughout the quarter, with a few exceptions listed below:
        - AI should <u>not</u> be used to provide responses to quizzes during the lecture.
        - AI should <u>not</u> be used to write reports automatically.
    - When interacting with these tools, especially in the context of large language models and chatbots derived thereof, students are *strongly encouraged* to fact-check all information produced. Models are prone to hallucination and factual information is often erroneous.
- The correct use of references is essential for this course. Remember what is stated by the UCLA Student Conduct Code: "*Unless otherwise specified by the faculty member, all submissions, whether in draft or final form, to meet course requirements (including a paper, project, exam, computer program, oral presentation, or other work) must either be the Student's own work, or must clearly acknowledge the source.*"
- All HWs are individual. Students are encouraged to discuss assignments and help each other debug code when necessary. However, all are expected to write their own results and computer code. If your solution is found to be unreasonably similar to other person's code, we will contact you and investigate the issue.

- Every part of this class, from the lectures to the office hours, is a space where all should be treated with respect. No form of harassment, discrimination, or violence is tolerated. For more information, refer to the Conduct Code and the broader UCLA Policy.
- If you have a preferred name that is not specified in your UCLA profile, feel free to let the instructor know so we can address you by your preferred name.

Other Comments

- This course of Machine Learning for Materials Science is a course with a lot of cutting-edge content on the field. New lectures, topics, and discussions may emerge during the quarter to accommodate the newest publications, achievements of AI, code and so on.
- The instructor will provide at least one paper or reading assignment per lecture to give additional examples and applications in a practical setting. Reading the paper is optional, but it can be an important way to learn more about the subject.
- Our lectures are about 2 hours long. In each lecture, we will try to have a 5-min intermission in the middle of the lecture for bio-breaks, stretching, and decompressing. Students are welcome to ask questions during this time too, and we will start our activities again 5 minutes after the beginning of the break.
- In principle, we will not have recorded lectures this quarter. If, for any reason, that changes, the instructor will upload the recorded lectures to BruinLearn prior to the actual lecture day. You are welcome to watch at your own pace.
- Last-minute changes to the lecture topics may happen. If that happens, we will update the schedule on lectures so you are always on top of everything.
- If you have constructive feedback about the course, lectures, homeworks, and other assignments, you are welcome to provide your thoughts during the quarter. Your feedback will be carefully considered to improve the course.